

SVM Compute Core — Full-Chip Design Summary (m5/v9: Final Harden)

Project: Multi-Class Cardiac Arrhythmia Detection — Caravel chipIgnite Tape-Out **Technology:** sky130A / sky130_fd_sc_hd **Flow:** OpenLane 2 v2.3.10 Classic **Architecture:** Batch v8/v9 — host pre-loads SV + input matrix; ASIC classifies autonomously **RTL freeze:** m5/rt1 v9 — NUM_SV=500, alpha_addr[8:0], reg_alpha_wr[24:0]

Component Summary

svm_compute_core (job 91966, v9)

Metric	Value
Clock	40 MHz (25 ns), TT corner clean
Setup WNS (TT)	+7.83 ns — 0 violations
Hold WNS (TT)	+0.30 ns — 0 violations
DRC	0 violations
Active power	~66 mW
Inference time	3.23 ms / beat (500 SVs × 256 dim at 40 MHz)
Avg power (80 bpm)	0.284 mW (0.431% duty cycle)
Die area	2500 × 2500 μm (~14% utilization)
GDS	226 MB
LEF	94 KB
GL netlist	13 MB

user_project_wrapper (job 91967)

Metric	Value
Die area	2920 × 3520 μm (Caravel fixed, FP_DEF_TEMPLATE)
Macro	u_svm at (253, 554) N — 2500 × 2500 μm footprint
Clock	wb_clk_i (Caravel), gated to svm_gclk via ICG
CTS	Disabled (RUN_CTS: 0)
DRC	11,923 Magic DRC (boundary artifacts — acceptable)
LVS	1,683 errors (boundary artifacts — acceptable)
GDS	230 MB
LEF	195 KB
GL netlist	78 KB

Functional Results

Implementation	Accuracy	SVs	Notes
sklearn OVR (float)	97.67%	416 total (unlimited)	float precision
ASIC binary OVR (Q6.10)	97.67%	500 total (100×5)	gamma=0.25, C=1.0

Zero accuracy gap — ASIC exactly matches sklearn on all 300 test samples. See Testing Set section for per-class breakdown.

Training Set

Parameter	Value
Source	MIT-BIH Arrhythmia Database (PhysioNet)
Split	80% train / 20% test — stratified, <code>random_state=42</code>
Beats per class	240
Classes	Normal (N), PVC, AFib, VT, SVT
Total training beats	1,200
Feature dimension	256 (128 single-beat + 64 10-beat mean + 64 RR history)
Precision	Q6.10 fixed-point (16-bit signed)

Testing Set

Parameter	Value
Source	MIT-BIH Arrhythmia Database (PhysioNet) — held-out 20%
Beats per class	60
Total test beats	300
sklearn accuracy	97.67% (293/300)
ASIC accuracy	97.67% (293/300) — zero gap vs. float

Per-class test results:

Class	Correct	Accuracy
Normal (N)	60/60	100.0%
PVC	60/60	100.0%
AFib	60/60	100.0%
VT	56/60	93.3%
SVT	57/60	95.0%

Batch Architecture (v8/v9)

Off-chip RAM Bus

Signal	Pin	Direction	Description
<code>ram_addr[18:0]</code>	GPIO[28:10]	ASIC out	{row[10:0], col[7:0]}
<code>ram_ren</code>	GPIO[29]	ASIC out	Read strobe
<code>ram_rdata[15:0]</code>	LA[15:0]	Host in→ASIC	Data valid after <code>RAM_LATENCY</code> cycles

Address layout: rows 0..499 = SV matrix; rows 500..1499 = input matrix.

RAM_LATENCY parameter — configures the number of clock cycles between `ram_ren` assertion and valid `ram_rdata`. Default is 1 (same-cycle combinational SRAM model used in cosim). Set to 3 for the IS61WV51216 async SRAM (10 ns access at 40 MHz → 3-cycle pipeline). The core inserts wait states automatically; the host does not need to pad responses. Verified by `svm_ram_latency_tb.sv` (10-beat, LAT=3, PASS in ~208 cycles/beat).

Rationale for LAT=3 over LAT=2: The IS61WV51216 specifies 10 ns access time, which fits within one 25 ns clock cycle in theory. However LAT=3 is used to provide a safety margin for: (1) PCB trace propagation delays between the ASIC GPIO pins and the SRAM data pins, (2) input flip-flop setup time inside the ASIC, and (3) SRAM slowdown at elevated temperature and low voltage. LAT=2 may work on a bench at room temperature but risks intermittent failures in field conditions. The throughput penalty (9.7 ms vs 3.23 ms per beat) is acceptable given the 750 ms heartbeat window at 80 bpm.

What the Host Does

MCU (low-power, continuous)

1. Collect 1000 heartbeats (250 Hz ECG → feature extraction)
 2. Load SV matrix (500 SVs × 256 features) → SRAM rows 0..499
 3. Load input matrix (1000 beats × 256 features) → SRAM rows 500..1499
 4. Write NUM_SAMPLES = 1000, write NUM_SV_0-4 = 100 each
 5. Write alpha coefficients via ALPHA_WR (Wishbone 0x28)
 6. Fire CONTROL[start]
- ▼ ASIC takes over (timing shown for RAM_LATENCY=1 / RAM_LATENCY=3):
- └─ LOAD_INPUT per beat: 256 / 768 cycles (reads from SRAM rows 500+)
 - └─ COMPUTE_DIST per SV: 258 / 770 cycles (reads SV from SRAM rows 0-499)
 - └─ COMPUTE_KERNEL: ~18 cycles (Horner LUT exp approximation, LAT-independent)
 - └─ WRITE_CLASS: argmax+alpha → sample_rdy (IRQ[0]) per beat
 - last beat → done (IRQ[1])

Inference time scales linearly with RAM_LATENCY. At 40 MHz: - LAT=1 (cosim default): **3.23 ms / beat** (500 SVs × 256 dim) - LAT=3 (IS61WV51216 SRAM): **~9.7 ms / beat** — still well within the 750 ms heartbeat period

Wishbone Register Map (base 0x3000_0000)

Offset	Name	R/W	Description
+0x04	CONTROL	RW	[0]=start [1]=vbatt_ok [2]=vbatt_warn
+0x08	STATUS	RO	[0]=done [1]=error [5:2]=error_code
			[8:6]=class [9]=sample_rdy
+0x0C	NUM_SAMPLES	RW	[9:0] beats in batch (1-1000)
+0x10-+0x20	NUM_SV[0-4]	RW	[7:0] SVs per class (max 100 each)
+0x24	PARAM_WR	WO	[19]=en [18:16]=addr [15:0]=data (gamma, C, bias)
+0x28	ALPHA_WR	WO	[24:16]=sv_global_idx (9-bit)
			[15:0]=alpha Q6.10

Full-Chip Power Estimate

Subsystem	Active Power	Duty Cycle	Avg Power
svm_compute_core (batch)	66 mW	0.431% (3.23 ms / 750 ms)	0.284 mW
Caravel management SoC	~5 mW	~5%	~0.25 mW
ECG frontend (analog)	~0.5 mW	100%	0.5 mW
BLE (optional, logging)	~10 mW	~0.1%	~0.01 mW
Total estimated	—	—	~1.04 mW

Battery budget: 200 mAh @ 3.7V = 740 mWh $\rightarrow$ **740 mWh / 1.04 mW 711 hours (~29.6 days)**. 14-day target met with $2.1\times$ margin. SVM core alone: ~2606 hours (~108 days).

Caravel Submission Artifacts

Repository: github.com/adamleehandwerger/caravel_svm_project

GDS Release: v2.0-hardened (226 MB core / 230 MB wrapper)

Layout artifacts:

File	Size	Job	Status
gds/svm_compute_core.gds	226 MB	91966	
gds/user_project_wrapper.gds	230 MB	91967	
lef/svm_compute_core.lef	94 KB	91966	
lef/user_project_wrapper.lef	195 KB	91967	

Verilog artifacts:

File	Size	Job	Status
verilog/gl/svm_compute_core.v	13 MB	91966	
verilog/gl/user_project_wrapper.v	78 KB	91967	
verilog/rtl/svm_compute_core.sv	—	v9	
verilog/rtl/user_project_wrapper.sv	—	v9	

Acknowledgments

Place-and-route was performed on **Orca**, Portland State University’s high-performance computing cluster, using SLURM batch jobs with OpenLane 2 v2.3.10 inside a Singularity container. We thank the PSU Research Computing team for providing access to the GPU and CPU nodes that made the multi-hour OpenLane runs feasible.

Feature Extraction References

The 256-dim multi-scale feature vector follows established AAMI EC57 beat classification conventions:

Feature group	Dims	Reference
Single-beat morphology (± 64 samples, amplitude-norm)	128	de Chazal P, O’Dwyer M, Reilly RB. “Automatic classification of heartbeats using ECG morphology and heartbeat interval features.” <i>IEEE Trans Biomed Eng</i> 51(7):1196-206, 2004. DOI: 10.1109/TBME.2004.827359

Feature group	Dims	Reference
10-beat mean morphology template	64	de Chazal P, Reilly RB. “A patient-adapting heartbeat classifier using ECG morphology and heartbeat interval features.” <i>IEEE Trans Biomed Eng</i> 53(12):2535-43, 2006. DOI: 10.1109/TBME.2006.883802
RR-interval history (99 intervals → 64 pts, norm to NORMAL_RR=308 ms)	64	Llamedo M, Martínez JP. “Heartbeat classification using feature selection driven by database generalization criteria.” <i>IEEE Trans Biomed Eng</i> 58(3):616-25, 2011. DOI: 10.1109/TBME.2010.2068048

Standard: AAMI ANSI EC57:2012 — Performance Requirements for Ambulatory ECG Analysers.

Dataset: PhysioNet MIT-BIH Arrhythmia Database (Moody GB, Mark RG, 2001).

DOI: 10.13026/C2F305

Document version: m5/v9 · 2026-05-28 — RAM_LATENCY parameter added; cosim 97.67% = sklearn; Appendix A added; Appendix B added

Appendix A — Runtime Model Reload Procedure

The ASIC is fully runtime-reprogrammable. A new trained SVM model can be loaded without resetting the chip or re-synthesizing. The on-chip alpha table and the off-chip SRAM are simply overwritten in place.

What constitutes a “model”

Parameter	Location	Size
SV matrix ($500 \times 256 \times 16$ -bit features)	Off-chip SRAM, rows 0–499	256 KB
Alpha coefficients (500×16 -bit)	On-chip <code>alpha_table[]</code> registers	8 KB
Gamma (, Q6.10)	On-chip <code>gamma_reg</code> shadow register	2 bytes
SV counts per class (<code>NUM_SV[0–4]</code>)	On-chip Wishbone registers	5 bytes

Reload sequence

Perform these steps while the ASIC is idle (`STATUS[done]=1` or after reset). Do **not** fire `CONTROL[start]` until all steps are complete.

Step 1 — Write new SV matrix to off-chip SRAM

Write $500 \text{ rows} \times 256 \text{ columns}$ of Q6.10 feature values to SRAM address rows 0–499. Address encoding: `addr[18:0] = {sv_global_idx[10:0], feat_dim[7:0]}`. This is a bulk SRAM write from the MCU and does not involve the Wishbone bus.

```
for sv in range(500):
    for feat in range(256):
        sram_write(addr=(sv << 8) | feat, data=sv_matrix[sv][feat])
```

Step 2 — Write new alpha coefficients via ALPHA_WR

Write all 500 alpha values over Wishbone. Each write encodes the SV global index and the alpha value in a single 32-bit word:

```

WB base = 0x3000_0000
ALPHA_WR offset = 0x28

for idx in range(500):
    word = (idx << 16) | (alpha[idx] & 0xFFFF) # [24:16]=sv_idx, [15:0]=alpha Q6.10
    wishbone_write(WB_BASE + 0x28, word)

```

Step 3 — Update gamma via PARAM_WR

```

PARAM_WR offset = 0x24
# addr=0 is gamma register; [19]=en, [18:16]=addr, [15:0]=value
word = (1 << 19) | (0 << 16) | gamma_Q6_10
wishbone_write(WB_BASE + 0x24, word)

```

Step 4 — Update SV counts if changed

Only required if the new model has different SVs per class than the previous model.

```

NUM_SV offsets: class 0 = 0x10, class 1 = 0x14, ..., class 4 = 0x20
for c in range(5):
    wishbone_write(WB_BASE + 0x10 + c*4, num_sv[c])

```

Step 5 — Fire start

```

wishbone_write(WB_BASE + 0x04, 0x0B) # CONTROL: start=1, vbatt_ok=1, kern_ready=1

```

Constraints

- Total SV count must not exceed 500 (100 per class maximum). Models exceeding this require re-synthesis with updated NUM_SV parameters.
- Gamma must be representable in Q6.10 (range 0–63.999, resolution ~0.001).
- Alpha values must be representable in Q6.10 (signed, range –32 to +31.999).
- The reload can be performed between any two batches. The ASIC does not need to be held in reset during the write sequence — the alpha table and gamma register are shadow-registered and only take effect when start fires.
- Writing ALPHA_WR while STATUS[done]=0 (i.e., during active inference) has undefined behavior and must be avoided.

Appendix B — Alternative Design: Hospital-Grade Batch Classifier (28nm)

This appendix specifies an alternative architecture targeting continuous bedside cardiac monitoring in a hospital environment — no power or area constraints, 100-beat batch processing, GEMM-based matrix engine.

B.1 Design Goals

Parameter	Wearable (current)	Hospital (proposed)
Use case	Ambulatory patch	Bedside monitor / ICU
Batch size	1000 beats	100 beats
Latency requirement	< 750 ms/beat	< 5 s per 100-beat batch
Power budget	< 1 mW avg	No constraint
Area budget	Caravel die (6.25 mm ²)	No constraint
Process	sky130A (180nm equiv.)	TSMC 28nm HPC

B.2 Feature Set (unchanged)

The same 256-dimensional multi-scale feature vector is used:

Group	Dims	Content
Single-beat morphology	128	± 64 samples, amplitude-normalized
10-beat mean template	64	Average of preceding 10 beats
100-beat mean template	64	Average of preceding 100 beats (replaces RR-interval history)
Total	256	

The 100-beat template replaces the RR-interval history for hospital use, providing longer-term morphological context available in a continuous-monitoring environment.

B.3 Architecture

Compute engine: 32×32 systolic array

A 32×32 weight-stationary systolic array processes the pairwise distance matrix as a single GEMM operation rather than sequentially iterating over SVs.

The distance computation is restructured as:

$$\begin{aligned} D[i,j] &= ||X[i] - SV[j]||^2 \\ &= ||X[i]||^2 - 2 \cdot (X \cdot SV^T)[i,j] + ||SV[j]||^2 \end{aligned}$$

The dominant term $X \cdot SV^T$ is a $100 \times 256 \times 256 \times 500$ GEMM — 12.8M MACs, fully parallelized across the systolic array. The squared norm terms are precomputed in a single pass and broadcast.

Memory: on-chip SRAM (1 MB total)

Buffer	Size	Contents
Input matrix	51.2 KB	100 beats $\times$ 256 features $\times$ 16-bit Q6.10
SV matrix	256 KB	500 SVs $\times$ 256 features $\times$ 16-bit Q6.10
Distance matrix	200 KB	$100 \times 500 \times 32$ -bit intermediate
Kernel matrix	100 KB	$100 \times 500 \times 16$ -bit exp output
Alpha table	1 KB	500×16 -bit (same as current)
Score / output	2 KB	$100 \times 5 \times 32$ -bit class accumulators
Total	~610 KB	(rounded to 1 MB with ECC and overhead)

Clock and process

Parameter	Value
Process	TSMC 28nm HPC
Supply voltage	0.9 V
Clock	800 MHz
Peak compute	$1024 \text{ MACs/cycle} \times 800 \text{ MHz} = \mathbf{819 \text{ GOPS}}$
On-chip SRAM bandwidth	~100 GB/s

B.4 Performance

Per 100-beat batch:

Stage	Operations	Cycles	Time
Load input + SV to SRAM	—	—	one-time, DMA
GEMM: $X \cdot SV^T$ ($100 \times 256 \times 500$)	12.8M MACs	~12,500	15.6 μ s
Squared norms + broadcast	100K ops	~100	0.1 μ s
Exp LUT (100×500 evaluations)	50K	50,000	62.5 μ s
Alpha accumulation ($100 \times 500 \times 5$)	250K MACs	~250	0.3 μ s
Argmax (100×5)	500	~1	<0.1 μ s
Total per 100-beat batch		~63,000	~78 μs

Throughput: 100 beats / 78 μ s = **1,280,000 inf/s (1.28 M inf/s)**

Duty cycle at 80 bpm (100 beats every 75 s): 78 μ s / 75,000,000 μ s = **0.000104%**

B.5 Power

Component	Active	Standby
Systolic array (32×32 , 0.9V, 800 MHz)	~180 mW	~0
On-chip SRAM (1 MB active)	~80 mW	~3 mW leakage
Control logic, IO	~40 mW	~0.5 mW
Total	~300 mW	~3.5 mW

Average power at 80 bpm (100-beat batches):

$$\begin{aligned} P_{\text{avg}} &= P_{\text{active}} \times \text{duty_cycle} + P_{\text{leakage}} \\ &= 300 \text{ mW} \times 0.000104\% + 3.5 \text{ mW} \\ &\approx 0.0003 \text{ mW} + 3.5 \text{ mW} \\ &= \sim 3.5 \text{ mW} \end{aligned}$$

Average power is dominated entirely by SRAM leakage. The compute contribution is negligible — the chip spends 99.9999% of the time idle.

B.6 Die Area

Block	Area (28nm est.)
1 MB on-chip SRAM	~0.50 mm ²
32×32 systolic array	~0.30 mm ²
Horner LUT + exp pipeline	~0.05 mm ²
Control FSM, IO, misc	~0.10 mm ²
Total estimated	~1.0 mm²

The hospital design is **6 $\times$ smaller die area** than the current sky130A core (6.25 mm²) despite being orders of magnitude more capable — a direct consequence of the 28nm process density advantage.

B.7 Roofline Comparison

The fundamental shift from the wearable to the hospital design is the operational intensity: the GEMM formulation reuses the SV matrix across all 100 input beats, dramatically increasing arithmetic intensity.

Operational intensity:

Design	Total ops	Memory traffic	Ops/byte
Wearable (sequential, per beat)	512K ops	256 KB (SV re-read each beat)	2.0
Hospital (GEMM, per 100-beat batch)	25.6M ops	307 KB (SV loaded once)	~83

The wearable design sits deep in the **memory-bound** region of the roofline — 98.9% of time waiting for SRAM data. The hospital design operates well above the ridge point (~16 ops/byte at 100 GB/s / 1.6 TOPS) and is firmly **compute-bound**.

B.8 Comparison Table

Throughput and latency:

Metric	Wearable ASIC	Hospital ASIC	sklearn	Numba (8-core)
Throughput	309 inf/s	1,280,000 inf/s	4,200 inf/s	95,000 inf/s
Latency / 100 beats	323 ms	0.078 ms	23.8 ms	1.05 ms
Speedup vs sklearn	0.07×	305×	1×	22.6×

Power, area and roofline:

Metric	Wearable ASIC	Hospital ASIC	sklearn	Numba (8-core)
Active power	66 mW	300 mW	15,000 mW	80,000 mW
Avg power (80 bpm)	0.284 mW	3.5 mW	~15,000 mW	~80,000 mW
Die area	6.25 mm ²	~1.0 mm ²	N/A	N/A
Process	sky130A (180nm)	TSMC 28nm	Intel 10nm	Intel 10nm
Ops/byte	2.0	83	~2.0	~2.0
Roofline regime	Memory-bound	Compute-bound	Memory-bound	Memory-bound

Key observations:

- The hospital ASIC is **305×** faster than sklearn and **13.5×** faster than 8-core Numba on a 100-beat batch
- This is achieved by converting a memory-bound sequential problem into a compute-bound GEMM
- The SRAM leakage floor (3.5 mW) makes the hospital design unsuitable for wearable use — 12×
- Both designs achieve 97.67% accuracy — the architecture difference is entirely in throughput and power, not classification quality